

Convex Optimization - Project 6

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a)

$$\min \frac{1}{T} \|y - S \star a\|_2^2 + \lambda r(a)$$
$$\text{s.t. } a \geq 0$$

For having sparsity in activation signal a , a good regularizer choice is $\ell_1(a)$ because it penalizes the abs sum of coefficients and forcing many of them to be zero.

$$r(a) = \|a\|_1 = \sum_{t=1}^N |a_t|$$

$$\stackrel{a \geq 0}{\Rightarrow} r(a) = \sum_{t=1}^N a_t$$

$$\Rightarrow \min \frac{1}{T} \|y - S \star a\|_2^2 + \lambda \sum_{t=1}^N a_t$$
$$\text{s.t. } a \geq 0$$

b, c) Code is attached.